

Quiz-1: Deep Learning : AI2100, AI5100 and CS5480

14th, February, 2026

Duration: 90 minutes

Total : 50 Marks.

Please read the following comments before you work on the problems:

- Please neatly write your name and roll number on the question paper, answer sheet, and every page of the extra sheet.
- Return the question paper and answer sheets once the exam ends.
- Please write the solutions neatly and clearly mention the corresponding question number. Answers written in illegible or cluttered handwriting may not be evaluated.
- This is a closed-book quiz. Calculators, phones, or any programmable devices are strictly prohibited. Any form of cheating or plagiarism, whether detected during the exam or later during grading, will result in **F grade in the course**.

1 Multiple Choice Questions (Single Correct)

[1 marks × 11 = 11 marks]

1. In the McCulloch–Pitts (MP) neuron model, what is the specific condition required for the neuron to fire (output 1)?
 - A. The weight of the inhibitory input must be more negative than the excitatory weights.
 - B. No inhibitory input must be active (ON).
 - C. The sum of inhibitory inputs must be less than the threshold θ .
 - D. All inhibitory inputs must be active simultaneously.
2. The gradient ∇f of a function $f : \mathbb{R}^D \rightarrow \mathbb{R}$ is a vector that points in which direction?
 - A. The direction of the fastest decrease of f .
 - B. The direction parallel to the contours of f .
 - C. A random direction that minimizes the loss.
 - D. The direction of the fastest increase of f .
3. Which mathematical identity correctly expresses the relationship between the sigmoid function $\sigma(z)$ and the hyperbolic tangent function $\tanh(z)$?

A. $\sigma(z) = \frac{\tanh(z)}{2}$ B. $\sigma(z) = \frac{1 + \tanh(z/2)}{2}$ C. $\sigma(z) = 1 + \tanh(z)$ D. $\sigma(z) = \tanh(z)$
4. In the context of machine learning theory, what is the primary reason for utilizing the empirical risk $\hat{R}(f; \mathcal{D})$ instead of the expected risk $R(f)$ during model training?
 - A. Empirical risk is guaranteed to be lower than expected risk, providing a more optimistic and efficient optimization target.
 - B. The expected risk only applies to regression problems, whereas empirical risk is versatile enough for classification.
 - C. The true data distribution is typically unknown, making the direct calculation of the average loss over all possible data points impossible.

- D. Empirical risk accounts for regularization terms, while expected risk is strictly the measure of the loss function without priors.
5. In a non-convex error surface, Batch Gradient Descent is guaranteed to converge to the global minimum if the learning rate is sufficiently small. Explain why.
A. True B. False
 6. Any Boolean function of n inputs must necessarily be represented with 2^n hidden perceptrons.
A. True B. False
 7. For linearly separable data with margin γ and bounded inputs $|x_i| \leq R$, the perceptron convergence theorem states that the number of mistakes M is bounded by:
A. $M \leq \frac{R}{\gamma}$ B. $M \leq \left(\frac{R}{\gamma}\right)^2$ C. $M \leq R^2\gamma$ D. $M \leq \frac{\gamma}{R}$
 8. In a perceptron, the decision boundary defined by $w^\top x + b = 0$ has which geometric interpretation?
A. A nonlinear manifold separating the input space.
B. A hyperplane whose normal vector is w .
C. A curved surface determined by the activation function.
D. A probabilistic boundary determined by softmax.
 9. If the activation function in a neural network is polynomial, then the network:
A. Is always a universal approximator.
B. Cannot approximate arbitrary continuous functions.
C. Becomes convex in parameters.
D. Always avoids vanishing gradients.
 10. Suppose we scale all weights and bias of a perceptron by a positive constant $\alpha > 0$. The decision boundary:
A. Rotates. B. Translates. C. Remains unchanged. D. Becomes nonlinear.
 11. Consider the perceptron loss $\ell(w; x, y) = \max(0, -y(w^\top x + b))$. When $y(w^\top x + b) > 0$, the gradient with respect to w is:
A. $-yx$ B. yx C. 0 D. Undefined

2 Multiple Choice Questions (Multi Correct)

[2 marks× 2=4 marks]

1. You have a deep neural network with 100 layers that takes images as input and outputs a number from $[0, 1]$ (representing how organized your desk is). However, you don't have enough computing resources to train all 100 layers from scratch. Fortunately, TiHAN has already trained the first 99 layers for you. You only need to train the last (100th) layer using mean squared error loss. Which of the following statements are TRUE?
 - A. If each layer uses the Tanh activation function, then TiHAN had to solve a non-convex optimization problem when training the first 99 layers.
 - B. You need to solve a non-convex problem because mean squared error loss is non-convex, even though the first 99 layers are already trained and frozen.
 - C. The problem you need to solve (training only the last layer) is convex.
 - D. If all layers use the identity activation function, then TiHAN's problem becomes convex because the entire network simplifies to a single linear layer.
 - E. If your desk images could be perfectly separated by a straight line/plane, then both your problem and TiHAN's problem would be convex.
2. Which of the following statements are FALSE?
 - A. The only way to train neural networks using first-order methods is via backpropagation because computing gradients naturally requires computing the chain rule.
 - B. The only way to train a neural network for an unsupervised learning problem is by assigning artificial labels to each data point and

converting it to a classification problem that we can solve. C. SGD is more computationally efficient than vanilla GD in practice. D. Multivariate polynomials are also not universal approximators because they can only model data that is linearly separable.

3 Write Short Answer / Fill in the blanks

[1 marks $\times$ 5=5 marks]

1. A neural network with a single hidden layer and a non-polynomial activation function can approximate any continuous function on a compact set according to the _____ theorem.
2. While the empirical risk is the average loss calculated specifically on the training set, the expected risk (or average risk) is defined as the average loss over the _____.
3. In the perceptron learning rule, when a misclassified example (x_i, y_i) is encountered, the weight update is $w \leftarrow w + \eta$, _____.
4. Consider a fully connected feedforward neural network with input dimension: 10, one hidden layer with 8 neurons, output layer with 3 neurons, and all layers include bias terms. Total number of parameters in the network is _____.
5. Consider a fully connected feedforward neural network with input dimension: 10, one hidden layer with 8 neurons, output layer with 3 neurons, and all layers include bias terms. Total number of parameters in the Hidden to Output layer for this network is _____.

4 Subjective Questions [Marks :30]

1. Why does Batch Gradient Descent become computationally “redundant” when dealing with very large datasets? Contrast this with how Stochastic Gradient Descent (SGD) addresses the issue. What are the drawbacks of stochastic gradient descent. [5]
2. Consider the binary classification data from Table 1. Perform perceptron training on this data clearly mentioning all the required symbols and steps (until a maximum of 5 epochs). [5]

sample (x_i)	label (y_i)
(1, 1, 1)	+1
(1, 1, 0)	-1
(1, 0, 1)	-1
(0, 1, 1)	-1

Table 1: Sample binary classification dataset.

3. Given the definition of the Tanh neuron as $\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$, derive its derivative $f'(z)$ in terms of $\tanh(z)$ itself. Use this result to justify why the gradient “stalls” for large values of z . [5]
4. You trained an MLP on MNIST hand-written digit classification task. Initially you planned to have 10 neurons in the output layer (one representing each of the classes starting from 0 to 9). However, later you wanted to create an output layer that predicts a binary representation of the digit. You decided to determine the bitwise representation of the input digit by adding an extra layer to the three-layer network above. The extra layer converts the output from the previous layer into a binary representation (MSB on the top), as illustrated in figure 1. Find a set of weights and biases (without training) for the new output layer. Assume that the first 3 layers of neurons are such that the correct output in the third layer (i.e., the old output layer) has activation at least 0.99, and incorrect outputs have activation less than 0.01. Define the activation function used in the new layer. [7.5]

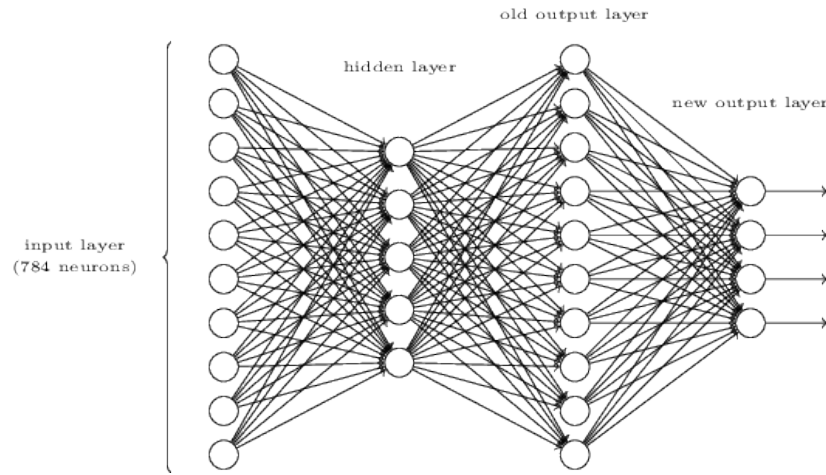


Figure 1: MLP for MNIST classification - Binary output

5. Consider the following feedforward neural network with three affine layers (four layers including the input), ReLU activations in the hidden layers, and three output classes.

Network architecture

- Input: $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \in \mathbb{R}^2$

- **Layer 1 (Affine + ReLU):**

$$z^{(1)} = W^{(1)}x + b^{(1)}, \quad a^{(1)} = \text{ReLU}(z^{(1)})$$

$$W^{(1)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad b^{(1)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- **Layer 2 (Affine + ReLU):**

$$z^{(2)} = W^{(2)}a^{(1)} + b^{(2)}, \quad a^{(2)} = \text{ReLU}(z^{(2)})$$

$$W^{(2)} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \quad b^{(2)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- **Layer 3 (Output layer, affine only):**

$$s = W^{(3)}a^{(2)} + b^{(3)}$$

$$W^{(3)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & -1 \end{bmatrix}, \quad b^{(3)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

There are three classes, and the true class label is $y = 1$ (using 1-based indexing, its the first class). The learning rate is $\eta = 0.1$.

Assume the derivative of the ReLU function is

$$\text{ReLU}'(t) = \begin{cases} 1, & t > 0 \\ 0, & t \leq 0 \end{cases}$$

Question: The multiclass hinge loss is defined as

$$L_{\text{hinge}} = \sum_{j \neq y} \max(0, s_j - s_y + 1)$$

1. Compute the gradient of the hinge loss with respect to the logits s . [2.5]
2. Using backpropagation, compute the gradients with respect to all weights and biases. [5]